Retrieval Augmented Generation

For Study-Aid Flashcards

Group 51

Ali Kömürcü – 12448700, Asad Zulfiqar – 12450744, Marvi Vani – 12536810,

Thomas Brezina – 11778916, Tom Pirich – 01525852

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1. Abstract

Students and researchers often struggle to efficiently synthesize dense academic documents into high-quality study materials without losing the ability to verify facts. This project aims to solve that problem by developing a Retrieval-Augmented Generation (RAG) system that transforms PDFs and notes into structured educational flashcards. The primary technical challenges involved implementing an optimal **Semantic Chunking** strategy, a **Hybrid Retrieval** architecture to ensure that retrieval is both conceptually and terminologically precise, and developing a multi-stage **Chain-of-Thought (CoT)** generation pipeline coupled with an **"LLM-as-a-Judge"** validation loop. This ensures that the generated output is not only pedagogically sound but strictly grounded in the source text through a verifiable citation framework, effectively mitigating hallucinations.

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2. Description

2.1 Ideal User Journey

Users upload multiple research papers or lecture notes in PDF, TXT, or Markdown formats. Once processed, the user can choose to query the documents via chat or generate a set of conceptual flashcards. These cards prioritize high-level "why" and "how" questions over rote memorization. To ensure transparency, both chat responses and flashcards include precise citations. This allows users to verify accuracy directly against the source text, transforming the AI from an opaque oracle into a verifiable learning guide.

2.2 Technical Approach and Reasoning

Our technical architecture prioritizes the quality of retrieved context, as the generation is only as good as the data it receives.

A. Ingestion and Semantic Chunking To ensure logical flow, we replaced fixed-size splitting with **Semantic Chunking** via `langchain_experimental`. Using the `nomic-embed-text` model, we apply a **95th percentile threshold** to trigger splits only at significant thematic transitions. This preserves the internal coherence of each chunk, which is vital for maintaining the context of educational content.

B. Hybrid Retrieval Engine Recognizing that neither vector nor keyword search is sufficient alone, we implemented a dual-stream engine:

- **Vector Retrieval:** *ChromaDB* with `nomic-embed-text` (768 dimensions) captures the conceptual "semantic meaning". Using *chroma*'s built in semantic search to find relevant documents.
- **Keyword Retrieval:** *BM25Okapi* with a regex tokenizer ensures precise matching of technical terms, acronyms, and formulas.
- **Reciprocal Rank Fusion (RRF):** We merge results by retrieving 3x the required candidates and reranking them using the formula:

  $$Score(d) = \sum_{r \in R} \frac{1}{60 + r(d)}$$

  This mathematical approach ensures that a document appearing high in *both* rankings (or exceptionally high in one) is prioritized, effectively balancing semantic relevance with exact term matching.

C. Multi-Stage Generation and Validation The system uses a specialized two-step **CoT** pipeline:

- **Extraction:** A prompt identifies key concepts strictly grounded in the retrieved chunks. Where each concept is strictly grounded in the source text.
- **Transformation:** A second prompt converts these into a structured set of cards based on pedagogical rules.
- **Self-Correction:** An **LLM-as-a-Judge** agent evaluates the cards on Accuracy, Completeness, Clarity, and Relevance (Scale 1-5). Only items with an average score ≥ 4 are shown to the user.

  Note on Reliability: We use a **low temperature** during the generation phase to limit the model's variance. This ensures the output remains focused and factual, significantly reducing the risk of hallucinations.

2.3 Challenges and Technical Obstacles

The development process revealed several critical hurdles in retrieval and generation:

- **Retrieval Evolution:** Initial keyword-only matching failed conceptual queries, while subsequent vector-only searches missed specific technical terms. Transitioning to a **Hybrid Search + RRF** approach was the breakthrough that balanced both needs.
- **Model Optimization:** We tested various embedding models, ultimately selecting `nomic-embed-text` for its superior "semantic-per-parameter" ratio, making it ideal for efficient local deployment.

- **The Citation Challenge**: Dense prompts initially caused "citation collapse," where the model defaulted to a single ID (e.g., `[1]`). While much improved through intensive, iterative prompt engineering to enforce grounding in the *`citation_id`* metadata, this remains a fragile area requiring further refinement for 100% consistency.
- **Schema Reliability:** Standard prompting yielded inconsistent JSON, breaking the pipeline. We resolved this by migrating to *GPT-4o-mini*, utilizing its native structured outputs capability to ensure strict schema adherence during generation, guaranteeing valid, machine-parseable outputs for our pipeline.

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3. Evaluation

The system was evaluated using 11 test cases: 8 domain-specific queries and 3 negative tests (out-of-scope) across three documents.

Results

- **Recall@5:** 86.5% (The system finds the correct context in the top 5 results most of the time).
- **MRR (Mean Reciprocal Rank):** 79.2% (The correct answer is typically in the first or second position).
- **Keyword Recall:** 89.9% (A heuristic measure of semantic accuracy).
- **Negative Testing:** 100% success rate in rejecting out-of-scope queries.

Discussion

The results indicate that the hybrid retrieval strategy is highly effective for academic data. The high Recall@5 suggests that the RRF algorithm successfully surfaces the most relevant information. Some limitations apply: the small sample size restricts generalizability, Keyword Recall offers only a heuristic approximation of semantic accuracy, and negative-case detection relies on phrase matching. The system occasionally struggles with citations in extremely dense documents, but overall, it produces contextually appropriate and pedagogical answers.

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4. Reflection

4.1 Human Agency

The system increases human agency. By automating the extraction and formatting of study materials, it frees the user to focus on higher-level synthesis and critical thinking. Because the system includes a robust citation and validation loop, it does not "replace" the student's judgment; rather, it provides the tools for the student to verify the AI's work, reinforcing a healthy, skeptical, and active learning relationship.

4.2 Further Development

Maybe. To avoid duplicating existing tools like NotebookLM, future development would focus on niche features like storytelling-based explanations or tracking user learning progress. Technically, we would explore advanced reranking methods to further optimize retrieval precision beyond the current RRF implementation.